MILITARY STANDARD

DESIGNATIONS
FOR
ELECTRIC SWITCHGEAR
AND
CONTROL DEVICES

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Department of Defense

OFFICE OF THE ASSISTANT SECRETARY OF DEFENSE

Supply and Logistics

WASHINGTON 25, D. C.

Designations for Electric
Switchgear and Control Devices
MIL-STD-27

1. This standard has been approved by the Department of Defense for use by the Departments of the Army, the Navy, and the Air Force.

2. In accordance with established procedure, the Office of Standardization has designated the Corps of Engineers, the Bureau of Yards and Docks, and the Air Force, respectively, as Army-Navy-Air Force custodians of this standard.

3. This standard is mandatory for use effective 1 January 1954 by the Departments of the Army, the Navy, and the Air Force.

4. Recommended corrections, additions, or deletions should be addressed to the Chief, Office of Standardization, Office of the Assistant Secretary of Defense (Supply and Logistics), Washington 25, D. C.

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1. GENERAL

1.1 Scope.—This standard covers designations that are used to identify the function performed by electric devices, and unit groups or assemblies of such devices, that are component elements of a complete control system or equipment. Switchgear device designations and control device designations are included.

1.2 Purpose.—The purpose of this standard is to integrate the essential elements of industry standards on similar subjects with the requirements of the Department of Defense in order to provide a basis for more accurate interpretation of the information shown on drawings and diagrams prepared by or for military activities.

2. DEFINITIONS

2.1 General.—Designations provide a means of identifying parts, assemblies of parts, and the functions of parts and assemblies, on drawings and diagrams. Generally the designations appearing on drawings are also marked on the items described, or adjacent to the items, on equipment to which they are attached. Associated parts lists and technical manuals use the designations as a ready means of reference to the items depicted on drawings.

2.2 Switchgear designations.—Switchgear designations are numbers that describe the functions performed by electric devices in switching circuits used in the generation, transmission, and distribution of electric power and in the utilization of electric power where the use of switchgear designations is desirable. Supplementary letters and numbers are used with the basic designation to permit positive identification of an item.

2.3 Control device designations.—Control device designations are letters used to identify the function of electric control devices, other than those used in aircraft, on power utilization equipment where such functional designations are required. Suitable prefix numbers and letters are added to the basic designation to discriminate between devices performing similar functions.

3. APPLICATION

3.1 General.—Switchgear and control device designations will apply as outlined under definitions. In general, on one complete drawing or set of drawings, only one system of designations shall be used.

3.2 Names.—Descriptions in the lists of designations contained in this standard include colloquial names as well as standard item names to facilitate identification. Standard item names are listed and defined in Part I of the Federal Item Identification Guides for Supply Cataloging.

3.3 On an individual equipment, supplementary letters or numbers shall have one meaning only. The meaning shall be clearly designated in the device list of the drawing. Where necessary to avoid conflict, other words beginning with the same letter shall be written out each time.

3.4 Exception to applications.—This standard is not applicable to designations used in electronic circuits, in electric circuits where functional designations are not suitable, and in aircraft electric circuits (see MIL-STD-16).

4. INDUSTRY STANDARDS

4.1 Switchgear designations.—The switchgear designations contained in this standard (formerly included in Military Standard JAN-STD-15, Electrical and Electronic Symbols) are in close agreement with the designations contained in the American Standards Association Standard for Automatic Station Control, Supervisory and Telemetering Equipment, C37.2-1945 and the National Electrical Manufacturers Association Standard for Power Switchgear Assemblies, SG5-1950.

4.2 Control device designations.—The control device designations contained in this standard are in close agreement with the National Electrical Manufacturers Association Standard IC1-23.2

5. SWITCHGEAR DESIGNATIONS

5.1 Numerical list.—The following is a list of switchgear designations arranged numerically:

1. Master Element
2. Time Delay Starting or Closing Relay
4. Master Circuit Breaker or Relay (Contactor)
5. Stopping Device

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6. Starting Circuit Breaker, Relay (Contactor), or Switch
7. Anode Circuit Breaker
8. Control Power Switch
9. Reversing Device
10. Unit Sequence Switch
11. Control Power Transformer
12. Overspeed Device
13. Synchronous Speed Device
14. Underspeed Device
15. Speed Regulating Device
16. Battery-Charging Control Device
17. Series Field Shunting Circuit Breaker or Relay (Contactor)
18. Accelerating or Decelerating Circuit Breaker or Relay (Contactor)
19. Starting to Running Transition Relay (Contactor)
20. Electrically-Operated Valve
21. Impedance Relay
22. Equalizer Circuit Breaker or Relay (Contactor)
23. Temperature Regulating Device
24. Bus Tie Circuit Breaker, Relay (Contactor), or Switch
25. Synchronizing or Paralleling Device
26. Apparatus Thermal Device
27. AC Undervoltage Relay or Device
28. Resistor Thermal Device
29. Isolating Circuit Breaker, Relay (Contactor), or Switch
30. Annunciator Relay
31. Separate Excitation Device
32. DC Reverse Power Relay
33. Position Switch
34. Motor-Operated Sequence Switch
35. Brush-Operating or Slip-Ring Short-Circuiting Device
36. Polarity Device
37. Undercurrent or Underpower Relay
38. Bearing Thermal Device
39. Field Reducing Relay (Contactor)
40. Field Relay
41. Field Circuit Breaker, Relay (Contactor), or Switch
42. Running Circuit Breaker, Relay (Contactor), or Switch
43. Transfer Device
44. Unit Sequence Starting Relay (Contactor)
45. DC Overvoltage Relay
46. Reverse-Phase, Phase-Balance Current, or Power Rectifier Misfire Relay
47. Single- or Reverse-Phase Voltage Relay
48. Incomplete Sequence Relay
49. AC Thermal Relay or Device
50. Short Circuit Selective Relay or Device
51. AC Overcurrent Relay
52. AC Circuit Breaker
53. Exciter or Generator Relay
54. High-Speed Circuit Breaker
55. Power Factor Relay
56. Field Application Relay or Device
59. AC Overvoltage Relay
60. Voltage Balance Relay
61. Current Balance Relay
62. Time Delay Stopping or Opening Relay
63. Fluid Pressure, Level or Flow Relay
64. Ground Protective Relay
65. Governor
66. Notching Relay
67. AC Power Directional or Power Directional Overcurrent Relay
68. DC Thermal Relay or Device
69. Permissive Control Device
70. Electrically-Operated Rheostat
71. DC Line Emergency Circuit Breaker or Relay (Contactor)
72. DC Line Circuit Breaker or Relay (Contactor)
73. Load Resistor Circuit Breaker or Relay (Contactor)
74. Alarm Relay
75. Position Changing Mechanism
76. DC Overcurrent Relay
77. Impulse Transmitter
78. Phase Angle Measuring Relay
79. AC Reclosing Relay
80. DC Undervoltage Relay or Device
81. Frequency Device
82. DC Reclosing Relay
83. Selective Control Relay (Contactor)
84. Operating Mechanism
85. Carrier- or Pilot-Wire Receiver Relay
86. Locking-out Relay
87. Differential Current Relay
88. Auxiliary Motor or Motor Generator
89. Line Switch
90. Regulating Device
91. DC Voltage Directional Relay
92. DC Voltage and Current Directional Relay

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93. Field Changing Contactor or Relay
94. Tripping or Trip-Free Relay (Contactor)
95.
96.
97. Reserved for special application
98.
99.

5.2 Alphabetical list.—The following is a list of switchgear designations arranged alphabetically and accompanied by a general description of the item function:

Accelerating or decelerating circuit breaker or relay (Contactor).—A device used to close or cause the closing of circuits for bringing a machine up to speed or shutting it down (18).

Alarm relay.—A relay other than an annunciator relay used to operate directly or in connection with a visual or audible alarm (74).

Annunciator relay.—A nonautomatic reset device which gives a number of separate visual indications upon the functioning of protective devices and which may also be arranged to perform the lockout function (30).

Anode circuit breaker.—A device used in the anode circuits of a power rectifier primarily to interrupt the rectifier circuit on arc back (7).

Apparatus thermal device.—A device which functions at desired high or low or both operating temperature values or limits of the apparatus to which it is applied (26).

Auxiliary motor or motor generator.—A unit used for operating auxiliary equipment such as pumps and blowers or motor-generators supplying excitation (88).

Battery-charging control device.—The battery charger (motor-generator set, tube or plate-type rectifier) (16).

Bearing thermal device.—A device which functions upon excessive bearing temperature (38).

Brush-operating or slip ring short-circuiting device.—A device for raising, lowering, or shifting the brushes of a machine, or for short-circuiting the slip rings (35).

Bus tie circuit breaker, relay (contactor), or switch.—A device which serves to connect buses or bus sections together (24).

Carrier- or pilot-wire receiver relay.—A relay operated or restrained by a blocking signal used in connection with carrier current or dc pilot-wire fault directional relaying (85).

Circuit breaker, AC.—A circuit breaker whose principal function is to interrupt short circuit or fault currents in an ac circuit (52).

Control power switch.—A manually operated switch used for the express purpose of connecting and disconnecting the source of control power to and from the control bus or equipment (8).

Control power transformer.—A transformer which serves as the source of ac control power for operating ac devices (11).

Current balance relay.—A relay which operates on a difference in current input or output of two circuits (61).

Differential current relay.—A fault-detecting relay which functions on a differential current of a given percentage or amount (87).

Electrically-operated rheostat.—A rheostat is used to vary the resistance of a circuit in response to some means of electrical control (70).

Electrically-operated valve.—A solenoid- or motor-operated valve in a vacuum, air, oil or water line or one used for braking purposes (20).

Equalizer circuit breaker or relay (contactor).—A device which serves to make and break the equalizer or current-balancing connections for machine field or machine-voltage regulators in a multiple unit installation (22).

Exciter or generator relay.—A relay which forces the building up of a dc machine field during starting or functions when the machine voltage has built up to a given value or does both (53).

Field application device.—A device used to control automatically the application of ac motor field excitation at some predetermined point in the slip cycle (56).

Field changing relay (contactor).—A device which functions to change the value of excitation on a machine (93).

Field circuit breaker, relay, (contactor), or switch.—A device which functions to apply or remove the machine field excitation or do both (41).

Field reducing relay (contactor).—A contactor which inserts resistance in the field circuit of a machine or otherwise reduces its field excitation (39).

Field relay.—A relay which functions on a given value of machine field current (40).

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Fluid pressure, level or flow relay.—A relay which operates on a given value of fluid or gas pressure, or flow, or fluid level (63).

Frequency device.—A device which functions on a given frequency value—either under, over, or normal (81).

Governor.—The equipment which controls the gate or valve opening of a prime mover (65).

Ground protective relay.—A relay which functions on failure of the insulation of a machine, transformer, or other apparatus to ground, or on flashover of a dc machine to ground (64).

High-speed circuit breaker. A circuit breaker which starts to reduce the current in the main circuit in 0.01 second or less, after the occurrence of a dc overcurrent or an excessive rate of current rise (54).

Impedance relay.—A relay which functions when the circuit impedance increases or decreases to a predetermined value (21).

Impulse transmitter.—A unit used to generate and transmit impulses over a telemetering or pilot wire circuit to a remote indicating or receiving device (77).

Incomplete sequence relay.—A relay which returns the equipment to the normal or off position and locks it out if the normal starting, operating, or stopping sequence is not properly completed within a predetermined time (48).

Isolating circuit breaker, relay (contactor) or switch.—A device used for the express purpose of disconnecting one circuit from another for purposes of emergency operation, maintenance, or test (29).

Line circuit breaker or relay (contactor), DC.—A device used to close and interrupt a dc power circuit under normal conditions or emergency conditions (72).

Line emergency circuit breaker or relay (contactor), DC.—A device used to interrupt a dc power circuit under emergency conditions (71).

Line switch.—A disconnecting or isolating switch, in an ac or dc power circuit (89).

Load resistor circuit breaker or relay (contactor).—A device used to shunt or insert a step of load limiting, heating, shifting, or indicating resistance in a power circuit (73).

Locking-out relay.—A relay which functions to shut down and hold an equipment out of service on the occurrence of abnormal conditions (86).

Master circuit breaker or relay (contactor).—A device generally controlled by the master element (1), and the necessary permissive and protective devices which serve to make or break the necessary control circuits to place an equipment in operation under the desired conditions and to take it out of operation under abnormal conditions (4).

Master element.—The initiating device such as control switch, voltage relay or float switch which serves either directly or through such permissive devices as protective and time-delay relays to place an equipment in or out of operation (1).

Motor-operated sequence switch.—A switch which fixes the operating sequence of the major devices during starting or stopping, or during other sequential switching operations (34).

Notching relay.—A relay which functions to allow only a specified number of operations of a given device or equipment, or a specified number of successive operations within a given time of each other. It is also used to allow periodic energizing of a circuit (66).

Operating mechanism.—Electrical mechanism, including the operating motor, position switches, etc., for a tap changer, induction regulator or similar apparatus (84).

Overcurrent relay, AC.—A relay which functions when the current in an ac circuit exceeds a given value (51).

Overcurrent relay, DC.—A relay which functions when the current in a dc circuit exceeds a given value (76).

Overspeed device.—A direct-connected speed switch which functions on machine overspeed (12).

Overvoltage relay, AC.—A relay which functions on a given value of ac circuit overvoltage (59).

Overvoltage relay, DC.—A relay which functions on a given value of dc circuit overvoltage (45).

Permissive control device.—A device consisting generally of a two-position, manually operated switch which permits the closing of a circuit breaker or the placing of an equipment into

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operation in one position and in the other position prevents operating the circuit breaker or equipment (69).

Phase angle measuring relay.—A relay which functions at a predetermined phase angle between voltages or currents or between voltage and current (78).

Polarity device.—A device which operates or permits the operation of another device on a given polarity only (36).

Position changing mechanism.—A mechanism used for moving a removable circuit breaker unit to and from the connected, disconnected, and test positions (75).

Position switch.—A switch which makes or breaks contact when the main device or apparatus, which has no device function number, reaches a given position (33).

Power directional or power directional overcurrent relay, AC.—A relay which functions on a desired value of power flow in a given direction or on a desired value of overcurrent with ac power flow in a given direction (67).

Power factor relay.—A relay which operates on a given power factor in an ac circuit (55).

Reclosing relay, AC.—A relay controlling the reclosing and locking out of an ac circuit interrupter (79).

Reclosing relay, DC.—A relay controlling the closing and reclosing of a dc circuit interrupter, generally in response to load circuit conditions (82).

Regulating device.—A device which functions to regulate a quantity such as voltage, current or power at a certain value or between certain limits (90).

Resistor thermal device.—A device which functions upon excessive temperature of load indicating, limiting, shifting, etc., resistors (28).

Reverse-phase, phase-balance current, or power rectifier misfire relay.—A relay which functions on polyphase current of reverse phase sequence, or on polyphase unbalance, or on failure of one or more of power rectifier anodes to fire (46).

Reverse power relay, DC.—A relay which functions on a given value of dc reverse power (32).

Reversing device.—A device used to reverse the shunt field connections of a synchronous converter during the starting sequence or for other special reversing functions (9).

Running circuit breaker, relay (contactor) or switch.—A device whose principal function is to connect a machine to its source of running voltage (42).

Selective control relay (contactor).—A device which operates to select automatically between certain sources or conditions in an equipment (83).

Series field shunting circuit breaker or relay (contactor).—A device used to open and close a shunting circuit around an item such as a capacitor, reactor, machine field or armature (17).

Separate excitation device.—A device which connects a circuit such as a synchronous converter shunt field to a source of separate excitation during the starting sequence, or for energizing the excitation and ignition circuits of a power rectifier (31).

Short circuit selective relay or device.—A device which functions instantaneously on an excessive value of current, or an excessive rate of current rise, indicating a fault in the apparatus or circuit being protected (50).

Single- or reverse-phase voltage relay.—A relay which functions upon a given value of polyphase voltage of the desired phase sequence (47).

Speed regulating device.—A device which functions to hold the speed or frequency of a machine or system at a certain value or between certain limits (15).

Starting circuit breaker, relay (contactor) or switch.—A device whose principal function is to connect a machine to its source of starting voltage (6).

Starting to running transition relay (contactor).—A device which operates to cause the transfer of a machine from the starting to the running power connections (19).

Stopping device.—A device whose primary function is to place and hold an equipment out of operation (5).

Synchronizing or paralleling device.—A device which operates when two ac circuits are within the desired limits of voltage, frequency, and phase angle to permit or cause the paralleling of these two circuits (25).

Synchronous speed device.—A device which operates at approximately normal synchronous speed of a machine, such as a centrifugal-speed

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switch, a slip-frequency relay, a voltage relay, an undercurrent relay (13).

Temperature regulating device.—A device which functions to hold the temperature of a machine or other apparatus between certain limits (23).

Thermal relay or device, AC.—A device which functions when the temperature of an ac machine or apparatus exceeds a given value (49).

Thermal relay or device, DC.—A device which functions when the temperature of a dc machine or apparatus exceeds a given value (68).

Time delay starting or closing relay.—A relay which gives the desired amount of time delay between operations in an automatic sequence (2).

Time delay starting or closing relay.—A relay which serves in conjunction with the device which initiates the shutdown, stopping, or opening indication in an automatic sequence (62).

Transfer device.—A manually operated device which transfers the control circuits to modify the plan of operation of the switching equipment or of some of the devices (43).

Tripping or trip-free relay (contactor).—A device which functions to trip a circuit breaker, contactor, or equipment, or to prevent immediate reclosure of a circuit interrupter, in case it opens an overload, even though the original closing circuit is maintained (94).

Undercurrent or underpower relay.—A relay which functions on a given minimum of current or power flow (37).

Underspeed device.—A device which functions on machine speed below a desired value (14).

Undervoltage relay or device, AC.—A device which functions on a given value of single phase ac undervoltage (27).

Undervoltage relay or device, DC.—A device which functions on a given value of dc undervoltage (80).

Unit sequence starting relay (contactor).—A device which functions to start the next available unit in a multiple unit equipment on failure or nonavailability of the normally preceding unit (44).

Unit sequence switch.—A switch used to change the sequence of placing units in and out of service in multiple unit equipment (10).

Voltage and current directional relay, DC.—A relay which operates in one manner when the dc voltage across an open circuit breaker or contactor exceeds a given value in a given direction, and operates in the opposite manner when the current in the circuit, after the circuit breaker or contactor is closed, exceeds a given value in the opposite direction (92).

Voltage balance relay.—A relay which operates on a difference in voltage between two circuits (60).

Voltage directional relay, DC.—A relay which operates when the dc voltage across an open circuit breaker or contactor exceeds a given value in a given direction (91).

5.3 Feeder equipment.—A series of numbers similar to that of paragraph 5.1 but starting with 101 instead of 1 shall be used for feeder equipments.

5.4 Automatic supervisory equipment for machines.—A series of numbers similar to that of paragraph 5.1 but starting with 201 instead of 1 shall be used for automatic supervisory equipment used with machines.

5.5 Automatic supervisory equipment for feeders.—A series of numbers similar to that of paragraph 5.1 but starting with 301 instead of 1 shall be used for automatic supervisory equipment used with feeders.

5.6 Hand reset.—The term "hand reset" shall be added whenever it applies.

5.7 Auxiliary switch designations.—Auxiliary switches for such uses as in relays (contactors), circuit breakers, safety enclosed trucks, housings, and enclosures shall be designated as follows:

(a) Closed when device is in energized or operated position.

(b) Closed when device is in deenergized or nonoperated position.

(aa) Closed when operating mechanism of main device is in energized or operated position.

(bb) Closed when operating mechanism of main device is in deenergized or nonoperated position.

(e), (f), (h), (k) Special auxiliary switches other than a, b, aa, or bb.

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5.7.1 Removable circuit-breaker unit (a) or (b), is an auxiliary contact on safety enclosed truck which is closed, or open, when the unit is in the connected position.

5.7.2 Housing or enclosure (a) or (b), is an auxiliary contact on the housing or enclosure which is closed, or open, when the removable circuit-breaker is in the connected position.

5.7.3 Where several auxiliary switches are present on same device they shall be designated numerically 1, 2, 3, etc., when necessary.

5.7.4 Where auxiliary relay contacts are used in place of auxiliary switches, a relay energized through an (a) contact is designated as "CL" and one energized through a (b) contact as "OP".

5.7.5 On circuit breakers, an auxiliary switch is designated as (a) (usually shown on drawings as normally open) if it is open when the circuit breaker is open, or as (b) (usually shown on drawings as normally closed) if it is closed when the circuit breaker is open.

5.8 Additional switchgear functional designations.—Where further definition for switchgear functions is required, the following letter and letter combinations shall be used:

5.8.1 *Alphabetically by designation*

A	Air; automatic; accelerating
B	Blower
BB	Bucking bar
C	Cold; close; compensator; current
CC	Closing coil
CS	Control switch
D	Down
F	Frequency
H	High; hot
HC	Holding coil
IS	Inductive shunt
IT	Inverse Time
L	Left; lower; low; leading; load limit; lower operating coil
M	Manual; metering
P	Power
PF	Power factor
Q	Oil
R	Raise; right
S	Solenoid; synchronizing
T	Trailing; trip
TC	Trip coil
U	Upper operating coil
V	Vacuum; voltage
VAR	Reactive power
W	Water
X, Y, Z	Auxiliary devices

5.8.2 *Alphabetically by name*

Accelerating	A
Air	A
Automatic	A
Auxiliary devices	X, Y, Z
Blower	B
Bucking Bar	BB
Close	C
Closing Coil	CC
Cold	C
Compensator	C
Control Switch	CS
Current	C
Down	D
Frequency	F
High	H
Holding Coil	HC
Hot	H
Inductive shunt	IS
Inverse time	IT
Leading	L
Left	L
Load limit	L
Low	L
Lower	L
Lower Operating Coil	L
Manual	M
Metering	M
Oil	Q
Power	P
Power factor	PF
Raise	R
Reactive power	VAR
Right	R
Solenoid	S
Synchronizing	S
Trailing	T
Trip	T
Trip coil	TC
Upper operating coil	U
Vacuum	V
Voltage	V
Water	W

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5.9 Examples.

5.9.1 The sequence of operation of automatic switchgear shall be shown where applicable as illustrated below.

1. The contacts of overcurrent relay 51 would be marked 51.

2. Relays picked up by relay 51 would be marked 51X, 51Y, 51Z.

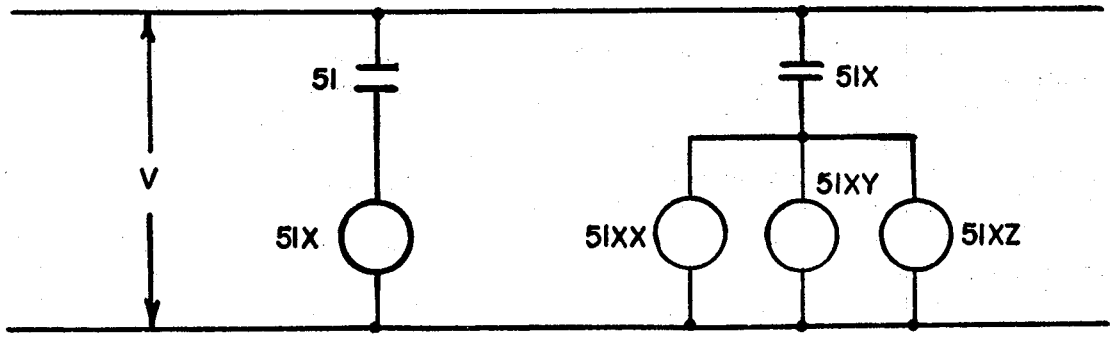


FIGURE 1.—*Example with one set of overcurrent relays.*

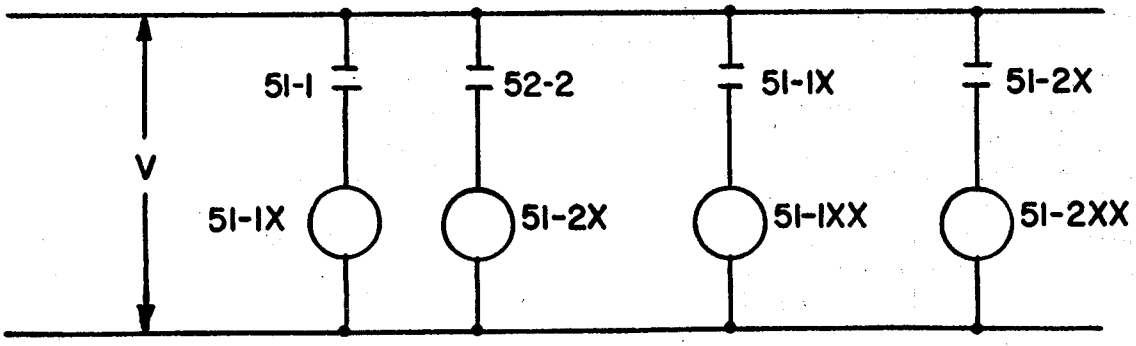


FIGURE 2.—*Example with two sets of overcurrent relays.*

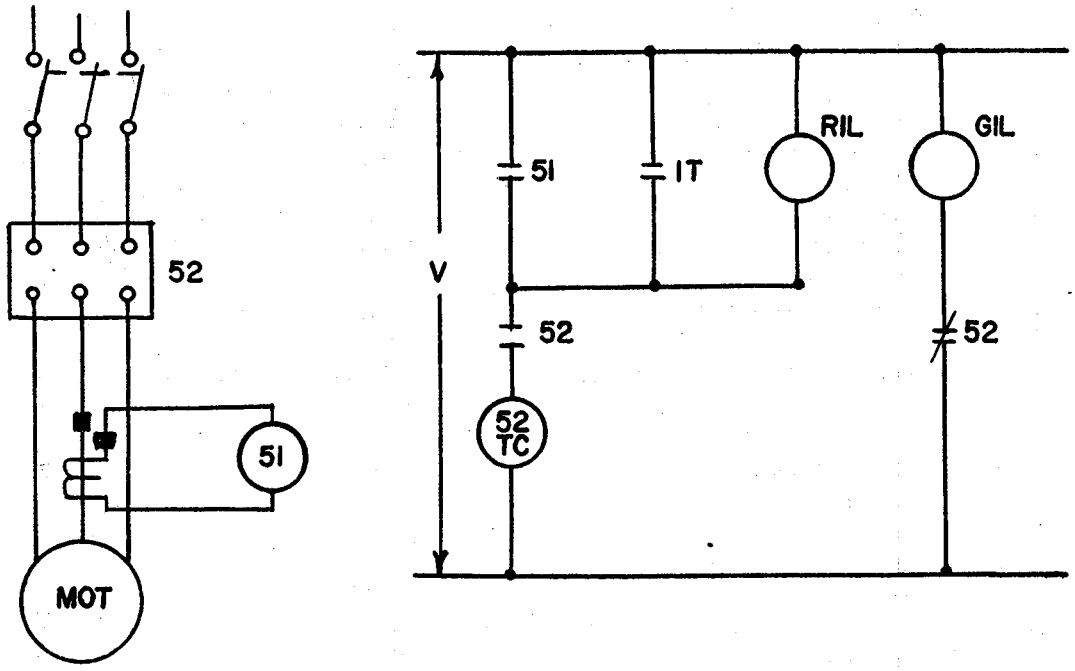


FIGURE 3.—*Example showing application of switchgear designations.*

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3. Relays picked up by relay 51X would be marked 51XX, 51XY, 51XZ.

4. If more than one 51 relay were used, the multiple relays would be identified as 51–1, 51–2, 51–3, etc.

5.9.2 One type of arrangement of the tripping circuit for a circuit breaker is shown in figure 3.

Where: 1T Master Element Tripping Contact
51 A-C Overcurrent Relay
52 A-C Circuit Breaker
52TC A-C Circuit Breaker Trip Coil
GIL Green Indicating Lamp
RIL Red Indicating Lamp
MOT Motor
V Voltage

1. With the circuit breaker 52 in the closed position the normally open contacts 52 are closed and RIL is illuminated. Further, normally closed contacts 52 are open and GIL is extinguished.

2. Overcurrent in the line to the motor will cause relay 51 to operate closing contacts 51 which completes the circuit to tripping coil 52 TC since normally open contacts 52 are closed. The circuit breaker opens and RIL is extinguished and GIL is illuminated because tripping of circuit breaker 52 closed normally closed contact 52.

3. If the circuit breaker is closed and it is desired to open it manually, operation of the Master Control to trip position will close contacts 1T and again RIL will be extinguished and GIL illuminated as the breaker trips by completion of the trip circuit.

6. CONTROL DEVICE DESIGNATIONS

6.1 Listing.—The following is a list of control device designations arranged alphabetically by designation and followed by general description of the item function:

Designation	*Function and Definition*
A	ACCELERATING AND DECELERATING CIRCUIT BREAKER, OR RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions for bringing a machine up to speed or shutting it down. (See designations DB, DE, EDE, ES, FD, and P)
AL	ALARM RELAY is a device other than an annunciator as covered under designation ANN, used to operate or to operate in conjunction with a visual or audible alarm.
AN	ANODE CIRCUIT BREAKER OR RELAY (CONTACTOR) is a device which serves to connect and disconnect a power rectifier from its source of ac power or to interrupt the rectifier circuit on arc back or to perform both functions.
ANN	ANNUNCIATOR RELAY is a nonautomatic reset device which gives a number of separate visual indications upon the functioning of protective devices and which may also be arranged to perform the lock-out function.
AS	ARMATURE SHUNT RELAY (CONTACTOR) is a device used to open and close a shunt circuit around a machine armature.
B	BRAKE RELAY (CONTACTOR) is a device which initiates the operation of an electrically operated brake.
BC	BALANCED CURRENT RELAY is a relay which operates on difference in current input or output of two circuits.
BG	BEARING TEMPERATURE RELAY is a relay which functions on excessive bearing temperatures.
BT	BUS-TIE CIRCUIT BREAKER RELAY (CONTACTOR) OR SWITCH serves to connect buses or bus sections together. (See designation PAR.)
BV	BALANCED-VOLTAGE RELAY is a relay which operates on a difference in voltage between two circuits.
CB	CIRCUIT BREAKER.
CL	CLOSE RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to operate a machine in the "close" direction.
CLA	CLAMP RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to perform a "clamp" operation.
CLU	CLUTCH RELAY (CONTACTOR) is a device used to initiate or cause the initiation of a circuit condition to operate a clutch.
CP	COMPELLING RELAY is a device which sequences or compels acceleration, usually of a multispeed ac machine through some lower speed connection before the machine may accelerate to a preselected higher speed.
CPT	CONTROL POWER TRANSFORMER is a transformer which serves as the source of ac control power for operating ac devices.
CSW	CONTROL POWER SWITCH is a manually operated switch used for the express purpose of connecting and disconnecting the source

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Designation	Function and Definition
	of control power to and from the control bus or equipment.
CUT	CUT RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to operate a machine to perform a "cut" operation.
D	DOWN RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to operate a machine to give mechanical motion in the "down" direction.
DB	DYNAMIC BRAKING RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to stop a machine by dynamic braking. (See designation A.)
DE	DECELERATING RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to stop a machine by regenerative braking or by some other method exclusive of dynamic braking, plugging and normal coasting stop. (See designation A.)
DIC	DIFFERENTIAL CURRENT RELAY is a fault-detecting relay which functions on a differential current of a given percentage or amount.
EDE	EMERGENCY DECELERATING RELAY (CONTACTOR) is a device used for emergency purposes to initiate or cause the initiation of circuit conditions to stop or reduce the speed of a machine. (See designation A.)
EQ	EQUALIZER CIRCUIT BREAKER OR RELAY (CONTACTOR) serves to make and break the equalizer or current-balancing connections for machine field or machine voltage regulators in a multiple-unit installation.
ES	EMERGENCY STOP DEVICE is a device used to initiate or cause the initiation of circuit conditions to produce emergency stopping of a machine or machines. For an individual equipment, it is necessary to define the significance of emergency stop. It may be any form of a braking stop or a coasting stop.
F	FORWARD RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to operate a machine to give mechanical motion in the forward direction.
FA	FIELD ACCELERATING RELAY (CONTACTOR) is a device which functions to force a dc machine field to accelerate the machine up to speed, in some cases as a function of certain limits of line current. (See designation FD)
FC	FIELD RELAY (CONTACTOR) is a device used to apply and remove excitation and discharge the field of a machine. (See designation FSW)
FD	FIELD DECELERATING RELAY (CONTACTOR) is a device which functions to force a dc machine field for the express purpose of slowing the machine down. (See designation FA)
FF	FULL-FIELD (FIELD-FORCING) OR FIELD-CHANGING RELAY (CONTACTOR) is a device which functions to change the value of excitation on a machine to obtain adequate torque for acceleration, for dynamic braking or for some other desired purpose. (See designations MT and LT)
FL	FIELD-LOSS RELAY (CONTACTOR) is a device which functions on a given value of machine field current for the purpose of detecting presence or loss of field current.
FP	FIELD-PROTECTIVE RELAY (CONTACTOR) is a device which inserts resistance in the field circuit of a machine or otherwise reduces field excitation while the machine is at rest for the purpose of protecting the field winding against overheating.
FQ	FREQUENCY DEVICE is a device which functions on a given value of frequency—either under, over or normal.
FR	FIELD-APPLICATION RELAY is a relay used to control automatically the application of machine field excitation.
FS	FAST-SPEED RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to operate a machine at the high (or maximum) speed.
FSW	FIELD SWITCH is a switch which is used to apply and remove excitation and discharge the field of a machine. (See designation FC)
FW	FIELD-WEAKENING OR FIELD-REDUCING RELAY (CONTACTOR) is a device which inserts resistance in the field circuit of a machine or otherwise reduces its field excitation. (See designation LT)
GP	GROUND-PROTECTIVE RELAY is a relay which functions on failure of the insulation of a machine, transformer or other apparatus to ground or on flashover of a dc machine to ground.
H	HOIST OR HOLD RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to operate a machine to give mechanical motion

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Designation	Function and Definition
	in the hoist direction (Hoist) or a device which functions by manual selection, through a master element (see designation ME), or the equivalent, (1) to initiate and maintain a desired condition (such as stall tension) prior to normal operation or (2) to hold and maintain an equipment in operation at the conditions prevailing when the selection is made and without interference from the automatic control (Hold).
IN	IN RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to operate a machine to give mechanical motion in the "in" direction.
IS	INCOMPLETE SEQUENCE RELAY is a relay which returns the equipment to the normal or "off" position and locks it out if the normal starting or operating or stopping sequence is not properly completed within a predetermined time.
ISW	ISOLATING CIRCUIT BREAKER RELAY (CONTACTOR) OR SWITCH is a device used for the express purpose of disconnecting one circuit from another for purposes of emergency operation, maintenance or test.
J	JOG RELAY (CONTACTOR) is a device which initiates or causes the initiation of circuits to permit intermittent acceleration or jogging of a machine at low speeds for mechanical positioning of the machine or the material being processed.
KO	KICK-OFF RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to operate a machine to "kick-off" material from a conveyor or roll-out table or to "kick-off" a crane hoist or for some similar purpose.
L	LOWERING OR LEFT RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to operate a machine to give mechanical motion in the lowering direction or in the left direction.
LC	LOAD-COMPENSATING RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to change or to adjust the field excitation of a machine or to otherwise provide compensation, usually as a function of load current, to maintain the controlled variable (voltage, speed, etc.) constant or within certain limits. (See designation RG.)
LO	LOCK-OUT RELAY OR DEVICE is a device which functions to shut down and hold an equipment out of service on the occurrence of abnormal conditions.
LPC	LOOP-CONTROL RELAY (CONTACTOR) is a device which initiates the operation of a machine to control the slack in a loop of material in a process line.
LR	LOAD-RESISTOR CIRCUIT BREAKER OR RELAY (CONTACTOR) is a device used to shunt or insert a step of load-limiting, heating, shifting or indicating resistance in a power circuit.
LS	LIMIT SWITCH is a switch used to detect a travel limit or any intermediate travel position for any form or adaptation of mechanical motion. This designation is usually expressed with variations as follows:
LSF	LIMIT SWITCH FORWARD
LSR	LIMIT SWITCH REVERSE
LSU	LIMIT SWITCH UP
LSD	LIMIT SWITCH DOWN
LSOP	LIMIT SWITCH OPEN
LSCL	LIMIT SWITCH CLOSE
LSW	LINE SWITCH is a disconnecting or isolating switch in an ac or dc power circuit.
LT	LOW-TORQUE RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to insert resistance in a dc machine field or armature or in the secondary circuit of a wound-rotor induction motor for the express purpose of reducing the machine torque to a pre-determined minimum value. (See designations FF and FW.)
M	MAIN-LINE RELAY (CONTACTOR) is a device whose principal function is to connect a machine or power circuit to its source of running or operating voltage.
ME	MASTER ELEMENT is the initiating device such as a control switch, voltage relay, pushbutton, float switch, etc., which serves either directly or through such permissive devices as protective and time delay relays to place an equipment in or out of operation. (See designations MS, CS and PB)
MR	MASTER RELAY (CONTACTOR) is a device generally controlled by a master element (see designation ME), or the equivalent, and the necessary permissive and protective devices, which serves to make and break the necessary control circuits to place an equipment in operation under the desired conditions and to take it out of operation under other or abnormal conditions.
MRH	MASTER RHEOSTAT OR ELECTRICALLY OPERATED RHEOSTAT is a rheostat used to vary or adjust the resistance or potential of a circuit in response to manual adjustment or to some means of electric control.

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Designation	Function and Definition
MS	MASTER SWITCH is usually a multiposition control switch. It functions to preset and initiate a unit-sequence for each of its operating handle position and is a special form of master element. (See designation ME)
MT	MAXIMUM-TORQUE RELAY (CONTACTOR) is a device which inserts or takes out resistance in a dc machine field or armature circuit in the secondary circuit of a wound-rotor induction motor for the express purpose of obtaining a machine torque of a certain maximum value. (See designation FF)
N	NEUTRAL RELAY (CONTACTOR) is a device which serves to open and close the neutral connection of a machine, transformer or circuit.
NO	NOTCHING RELAY is a relay which functions to allow only a specified number of operations of a given device or equipment, or a specified number of successive operations within a given time of each other. It is also used to allow periodic energizing of a circuit.
OC	OVERCURRENT RELAY is a relay which functions when the current in a circuit exceeds a given value.
OL	OVERLOAD RELAY is a relay which functions when the load of the machine or apparatus becomes excessive.
OP	OPEN RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to operate a machine to give mechanical motion in the open direction.
OPR	OPEN-PHASE, REVERSE-PHASE RELAY is a relay which functions when one phase of a polyphase circuit opens or when the phase sequence of a polyphase circuit reverses.
OS	OVERSPEED DEVICE is usually a direct connected speed switch which functions on machine overspeed.
OUT	OUT RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to operate a machine to give mechanical motion in the "out" direction.
OV	OVERVOLTAGE RELAY is a relay which functions on a given value of circuit overvoltage.
P	PLUGGING RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuits to stop or stop reverse a machine by plugging. (See designation A)
PAR	PARALLELING CIRCUIT BREAKER RELAY (CONTACTOR) OR SWITCH which connects two circuits or machines in parallel. (See designation BT)
PE	PERMISSIVE CONTROL DEVICE is generally a two-position manually operated switch which permits the closing of a circuit breaker or relay (contactor) or the placing of an equipment into operation in one postion and, in the other position, prevents operating the circuit breaker or relay (contactor) or equipment.
PF	POWER-FACTOR RELAY is a relay which operates on a given power factor in an ac circuit.
PS	PRESSURE SWITCH is a switch which operates on a given value of fluid pressure. This designation is usually used with a subscript designating the medium to which the pressure switch responds: i.e., air, oil, etc.
R	REVERSE RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to operate a machine in the "reverse" direction.
RA	RAISE RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to operate a machine in the "raise" direction.
RC	REVERSED-CURRENT (POWER) RELAY OR DEVICE, D C is a device which functions on a given value of dc reverse current (power).
RET	RETURN RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to operate a machine in the "return" direction.
RG	REGULATING DEVICE is a device which functions as a component of a regulating system to maintain or adjust a quantity such as voltage, current, power, etc., at a certain value or between certain limits.
RSW	REVERSING SWITCH is a manually operated device used to reverse a power circuit.
RT	RIGHT RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to operate a machine in the "right" direction.
RUN	RUN RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to connect a machine to its rated source of running voltage.
S	STARTING RELAY (CONTACTOR), CIRCUIT BREAKER OR SWITCH is a device whose principal function is to connect a machine to its source of starting voltage.
SC	SQUIRREL-CAGE RELAY is a protective device which functions to protect the squirrel-cage starting winding of a syn-

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Designation	Function and Definition
	chronous motor during starting or stalled conditions.
SD	SLOW-DOWN RELAY (CONTACTOR) is a device used to initiate or cause the initiation of a change in the operation of a machine from one speed to some lower operating speed.
SE	SELECTIVE CONTROL (OR TRANSFER) RELAY (CONTACTOR) is a device which operates to select between certain sources of conditions in an equipment or which performs automatically a transfer operation.
SR	SERIES CIRCUIT BREAKER RELAY (CONTACTOR) OR SWITCH serves to connect two circuits or machines in series.
SS	SLOW-SPEED RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to operate a machine at slow speed.
ST	STALL RELAY (CONTACTOR) is a device used to insert resistance in the field or armature circuit of a dc machine for the deliberate purpose of stalling a motor-drive unit.
STP	STOPPING RELAY is a relay whose primary function is to open a circuit breaker or relay (contactor) or to take an equipment out of operation.
SU	SUICIDE (NEUTRALIZING) RELAY (CONTACTOR) is a device which initiates or causes the initiation of circuit conditions to reconnect the field circuit of a machine, usually across the output terminals, for the express purpose of degenerating the output voltage to a minimum possible value and neutralizing the residual of the machine.
T	TORQUE RELAY (CONTACTOR) is a device which initiates a sequence to operate a machine to maintain torque.
TEN	TENSION RELAY (CONTACTOR) is a device which initiates a sequence to operate a machine or equipment to maintain tension of the material in a process line.
TH	THREAD RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to accelerate a machine or machines to a reduced operating speed for the purpose of threading or engaging the material being processed prior to normal operation.
TIL	TILT RELAY (CONTACTOR) is a device which initiates or causes the initiation of circuit conditions to perform a "tilt" operation.
TR	TIME-DELAY RELAY is a relay which gives a desired amount of time delay between operations in an automatic sequence.
TSW	TRANSFER (OR SELECTOR) SWITCH is a manually operated device which transfers the control or power circuits to modify the plan of operation of the switching equipment or of some of the devices.
U	UP RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to operate a machine to give mechanical motion in the "up" direction.
UC	UNDERCURRENT (UNDERPOWER) RELAY is a relay which functions on a given minimum value of current (underpower).
UCL	UNCLAMP RELAY (CONTACTOR) is a device used to initiate or cause the initiation of circuit conditions to operate a machine to perform an "unclamp" operation.
US	UNDERSPEED RELAY is a relay which functions on machine speed below a desired value.
UV	UNDERVOLTAGE RELAY is a relay which functions on a reduction or failure of dc voltage or single-phase ac voltage.
V	VOLTAGE RELAY is a relay which functions on a given value of voltage.
VLV	ELECTRICALLY-OPERATED VALVE is a solenoid or motor-operated valve in a vacuum, air, oil or water line, etc., or one used for braking purposes.

6.2 Miscellaneous control device designations.—The following designations shall be used to identify miscellaneous items:

AH	Anti-hunt or stabilizing transformer or device
AM	Ammeter
AMS	Ammeter transfer switch
ASX	Armature saturable reactor
C	Capacitor
CS	Control switch
CT	Current transformer
DM	Demand meter
DS	Door switch
FM	Frequency meter
FU	Fuse
GD	Ground detector
GND	Ground
IL	Indicating lamp (always using prefix letter indicating color: e. g. RIL-red lamp)
PB	Pushbutton
PS	Pressure switch
PT	Potential transformer
PFM	Power factor meter

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RAM	Recording ammeter
RDM	Recording demand meter
REC	Rectifier
RES	Resistor
RH	Rheostat
RTM	Recording tachometer
RVM	Recording voltmeter
SX	Saturable reactor
T	Transformer
TACH	Tachometer generator
TB	Terminal board
TM	Time or temperature meter
TS	Time switch
TT	Thermostat
TVM	Tachometer voltmeter
VAR	Varmeter
VARHM	Var-hour meter
VM	Voltmeter
VMS	Voltmeter transfer switch
WHM	Watt-hour meter
WM	Wattmeter
X	Reactor

6.3 Auxiliary devices.—The following suffixes may be used to denote auxiliary devices:

D	Down position, switch relay
F	Forward position, switch relay
R	Reverse position, switch relay
U	Up position, switch relay
X, Y, Z	Auxiliary relay or contact
CL	Closed position, switch relay
OP	Open position, switch relay

6.4 Parts of device.—The following subscripts may be used to denote parts of a device:

A or AUTO	Automatic
B	Brake
C	Current; capacitor
CC	Closing coil
CL	Closed; cold
D	Down
H	Hot, high
HC	Holding coil
HR	Hand reset
IT	Inverse time
L	Low; left; lower operating coil
LS	Limit switch
M	Operating motor
M or MAN	Manual
OFF	Off
P	Potential
PM	Polarity mark
SOL	Solenoid
SYN	Synchronizing
T	Test, trip
TC	Time-relay closing; trip coil
TD	Time delay
TO	Time-delay opening
U	Up; upper operating coil
UC	Unloader coil

6.5 Devices having same function.—Prefix numbers may be used to distinguish between two or more devices having the same function. The numbers shall be selected in numerical order starting with one. Where an automatic sequence is involved, the numbers shall be assigned in the order of the functional sequence.

For example:

1 SR, 2 SR, 3 SR, designate the first, second and third stopping relay and the order in which the sequence would be initiated in an automatic stopping sequence.

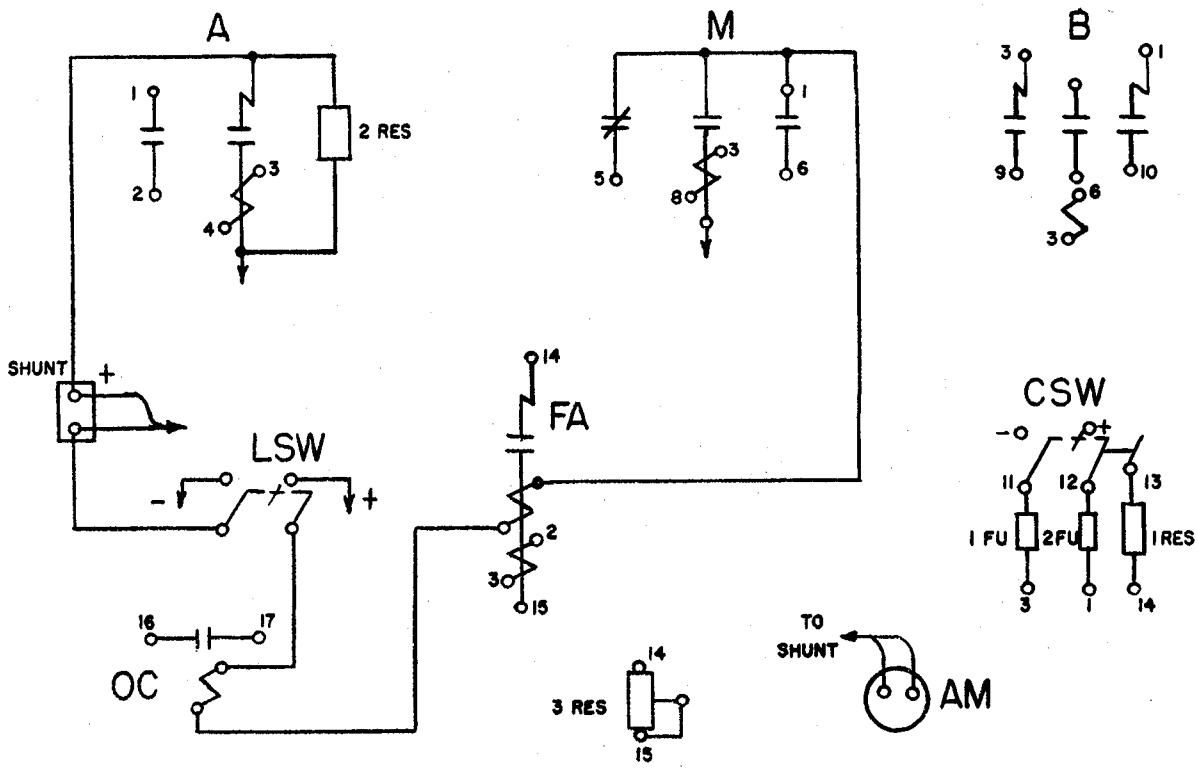
Custodians:

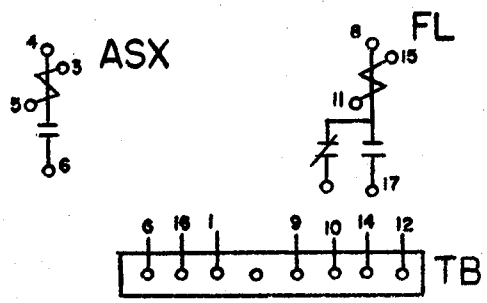
Army—Corps of Engineers
Navy—Bureau of Yards and Docks
Air Force

Other interest:

Army—CMOQ Sig T
Navy—*M*C Or Sh *CG*

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WIRE TABLE		
TERMINAL		CONNECT TO
+		LSW, CSW
-		LSW, CSW
1	TB	1FU, M, A, B
2		A, FA
3		1FU, M, A, B, ASX, FA
4		A, ASX
5		M, ASX
6	TB	M, ASX, B
8		M, FL
9	TB	B
10	TB	B
11		FL, CSW
12	TB	CSW
13		1RES, CSW
14	TB	1RES, 2 RES, FA
15		FA, 3RES, FL
16	TB	OC
17		OC, FL

FIGURE 4.—*Diagram illustrating use of control device designations.*

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